Vector+ *The 1-Bit Hardware Path*

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The Core Architecture

LatticeRunner is a neuromorphic associative memory engine operating a 4096×4096 lattice of binary neurons (16.7M cells). All operations — Hebbian learning, attractor settling, pattern recall — use **1-bit neurons, 1-bit weights, XOR, and popcount.** Zero floating-point arithmetic at the serving layer. The entire learned state of 1 million patterns fits in a **12 MB binary file** (brain cartridge) regardless of pattern count.

Operations are local (each neuron reads only neighbors and weights), deterministic (same input always produces same output), and fully parallelizable. This is the profile of a system that maps directly to systolic arrays, FPGA fabric, or custom ASIC — no branch predictors, no cache hierarchies, no floating-point units required.

Validated Results (February 2026)

Metric	Result	Dataset
Capacity (real embeddings, clean)	**R@1 = 1.000 at 1,000,000**	Wikipedia / Nomic 1.5
Capacity (real embeddings, 10% bitflip)	**R@1 = 1.000 at 1,000,000**	Wikipedia / Nomic 1.5
Capacity (real embeddings, 10% erasure)	**R@1 = 1.000 at 1,000,000**	Wikipedia / Nomic 1.5
Synthetic ceiling (no wall detected)	**R@1 = 1.000 at 3,000,000**	768-dim Gaussian
Training-side noise immunity	**R@1 = 1.000 at 50% noise**	All noise types
Quantization: float32 → 1-bit	**R@1 = 1.000 at every level**	128 MB → 12 MB, 0 recall loss
Bitpacked speed vs float32	**1.86x faster**	RTX 4080 Super

The Quantization Ladder: GPU → FPGA → ASIC

Each step validated before the next begins. The bitpacked (V9) representation *is* the hardware representation — there is no translation step between software and silicon.

Platform	Status	Brain Size	Operations	FP Required
GPU (V7 float32)	**Production**	128 MB	Float MACs	Yes
GPU (V8 INT8)	**Validated**	128 MB	INT8 dp4a	**No**
GPU (V9 1-bit)	**Validated**	**12 MB**	XOR + popcount	**No**
FPGA	Architecture ready	12 MB (BRAM)	LUTs + BRAM	**No**
ASIC	Paper design	12 MB (on-die)	Comb. logic	**No**

FPGA target: Lattice Semiconductor Avant-70. A 2048×2048 brain tile fits entirely in on-chip BRAM. The physics operations — stencil convolution, threshold comparison, popcount — are pure combinational logic. No multiply-accumulate units. No floating-point pipelines. No memory controllers for weight tensors. A 4×4 tile grid (4096×4096 total) can be networked on a single board.

The 1-bit no-learning variant ships as a 12 MB weight file. Training occurs offline on GPU (full-precision Hebbian learning with BCM homeostatic scaling). The trained weights are quantized to 1-bit and published to the serving tier. The serving device never trains — it only recalls.

The Endgame: A Memory Device

The ASIC target is not a general-purpose AI accelerator. It is a **memory device** — analogous to how DRAM is a storage device. Fixed function: imprint, settle, recall. Binary operations at fixed clock rates. Predictable power envelope. The TAM for dedicated AI associative memory hardware is a subset of the $50–150B datacenter infrastructure market as memory-intensive agentic architectures become standard.

Current Ask

$25,000–$50,000 angel funding to finalize benchmarks for arXiv publication, advance FPGA prototyping conversations, and complete the production software release. Active engagement with PSBA. Two partnership discussions underway with Portland-area IT services firms.

Next round: $250K pre-seed to begin FPGA prototyping, expand enterprise pilots, and advance the datacenter deployment roadmap.

IP: Core LatticeRunner technology held in dedicated IP entity. Provisional patent filing planned for full stack. Oregon-based — positioned within the Pacific Northwest hardware and datacenter ecosystem (Intel, Lattice Semiconductor, NVIDIA).